PCBC H Parametric & Sequences Test (Practice D) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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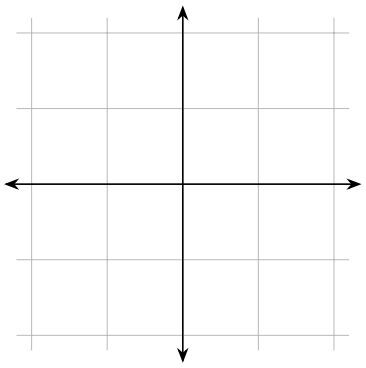
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Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

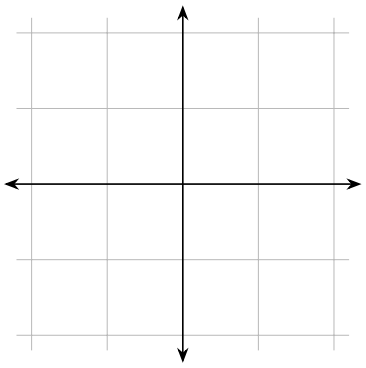
1. $x = -2t + 4, \quad y = \dfrac{3t^2}{2} - 6t + 6, \quad 0 \le t \le 4$

SIMPLIFIED rectangular form: _____



2. $x = t + 1, \quad y = \dfrac{4}{t+1}, \quad 1 \le t \le 3$

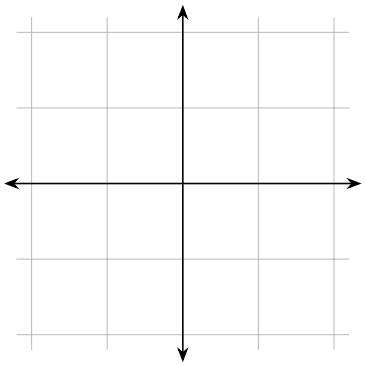
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

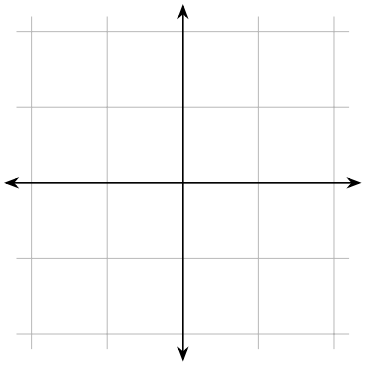
3. $x = 4\sin t, \quad y = 3\cos t, \quad 0 \le t \le \pi$

rectangular form: _____



4. $x = 3\tan t, \quad y = -2\sec t, \quad -\dfrac{\pi}{2} < t < \dfrac{\pi}{2}$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = \dfrac{1}{2}(x+3)^2 - 4$

Write out the sum. (2 points)

6. $\displaystyle\sum_{k=0}^{n} (-1)^k \frac{k+1}{2^k}$

Extra practice: the other sequence questions #6 could be

Not part of the 16 points. One of these types is the one sequence question; worked in the key.

Write down the first five terms of the sequence.

E1. $\{a_n\} = \left\{(-1)^n \dfrac{2^n}{2n+1}\right\}$

A sequence is defined recursively. Write the first five terms.

E2. $a_1 = 1; \quad a_2 = 2; \quad a_n = a_{n-1} - a_{n-2}$

Express the sum using summation notation.

E3. $\dfrac{3}{4} - \dfrac{4}{9} + \dfrac{5}{16} - \cdots + \dfrac{21}{400}$

Answer Key

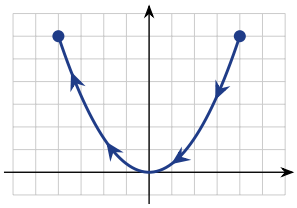
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = -2t + 4,\ y = \frac{3t^2}{2} - 6t + 6,\ 0 \le t \le 4$

t	(x, y)
0	$(4, 6)$
1	$(2, \frac{3}{2})$
2	$(0, 0)$
3	$(-2, \frac{3}{2})$
4	$(-4, 6)$

brackets: plug in both ends



starts at $(4,6)$ ($t = 0$) and moves right to left; solid dots, no end arrows

$x - 4 = -2t \ \Rightarrow\ t = \frac{4-x}{2}$ — *isolate t in x: one answer; divide by -2*

$y = \frac{3}{2}\left(\frac{4-x}{2}\right)^2 - 6\left(\frac{4-x}{2}\right) + 6$ — *substitute*

$y = \frac{3}{2} \cdot \frac{16 - 8x + x^2}{4} - 3(4 - x) + 6$ — *square the whole fraction: bottom $2^2 = 4$*

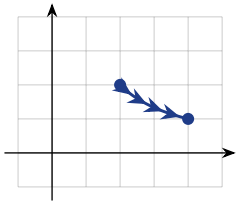
$y = \frac{3}{8}x^2 - 3x + 6 - 12 + 3x + 6$ — *distribute $\frac{3}{8}$ and -3*

$\boxed{y = \frac{3}{8}x^2 \text{ for } x \text{ in } [-4, 4]}$ — *SIMPLIFIED: y alone; domain from the t-chart*

Check a t-chart point: $x = 2$ gives $\frac{3}{8}(4) = \frac{3}{2}$. Matches.

2. $x = t + 1,\ y = \frac{4}{t+1},\ 1 \le t \le 3$

t	(x, y)
1	$(2, 2)$
2	$(3, \frac{4}{3})$
3	$(4, 1)$



left to right; solid dots

$t = x - 1$ — *isolate t*

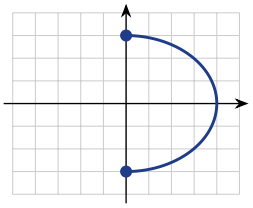
$y = \frac{4}{(x-1)+1}$ — *substitute*

$\boxed{y = \frac{4}{x} \text{ for } x \text{ in } [2, 4]}$ — *only a piece of the curve: give the domain*

Without an interval, $y = \frac{4}{x}$ alone is right; $x \neq 0$ comes with the equation.

3. $x = 4\sin t,\ y = 3\cos t,\ 0 \le t \le \pi$

t	(x,y)
0	$(0,3)$
$\frac{\pi}{2}$	$(4,0)$
π	$(0,-3)$



right half only; direction not asked

$\sin t = \frac{x}{4},\quad \cos t = \frac{y}{3}$

$\frac{x^2}{16} + \frac{y^2}{9} = 1$ $\qquad$ $\sin^2 t + \cos^2 t = 1$

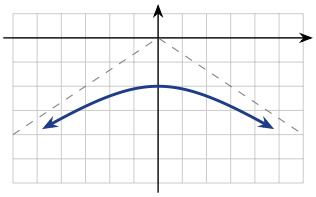
$\boxed{\frac{x^2}{16} + \frac{y^2}{9} = 1 \text{ for } x \text{ in } [0,4]}$ $\qquad$ *only the right half is traced; a domain cuts it*

Half the ellipse, not the whole thing: $x = 4\sin t \ge 0$ on $[0,\pi]$.

4. $x = 3\tan t,\ y = -2\sec t,\ -\frac{\pi}{2} < t < \frac{\pi}{2}$

t	(x,y)
$-\frac{\pi}{4}$	$(-3,-2\sqrt{2})$
0	$(0,-2)$
$\frac{\pi}{4}$	$(3,-2\sqrt{2})$

parentheses: ends undefined



bottom branch only; end arrowheads: it keeps going; asymptotes optional

$\tan t = \frac{x}{3},\quad \sec t = -\frac{y}{2}$

$\frac{y^2}{4} - \frac{x^2}{9} = 1$ $\qquad$ $\sec^2 t - \tan^2 t = 1$

$y^2 = 4\left(\frac{x^2}{9} + 1\right)$ $\qquad$ *bottom branch only; domain can't cut it: solve for y*

$\boxed{y = -2\sqrt{\frac{x^2}{9} + 1}}$ $\qquad$ *keep the $-$ sign; never "$y \le -2$"*

5. $y = \frac{1}{2}(x+3)^2 - 4$

$\boxed{x = t,\ y = \frac{1}{2}(t+3)^2 - 4}$

$x = t - 3:\ y = \frac{1}{2}(t - 3 + 3)^2 - 4$ $\qquad$ *pick $t-3$ so the inside is t*

$\boxed{x = t - 3,\ y = \frac{1}{2}t^2 - 4}$

Both for t in $(-\infty,\infty)$ (assumed; writing it is fine). Give both $x =$ and $y =$. Don't expand.

6.

$(-1)^0 \frac{1}{2^0} + (-1)^1 \frac{2}{2^1} + (-1)^2 \frac{3}{2^2} + \cdots + (-1)^n \frac{n+1}{2^n}$ $\qquad$ *start at $k = 0$, not 1*

$\boxed{1 - 1 + \frac{3}{4} - \cdots + (-1)^n \frac{n+1}{2^n}}$ $\qquad$ *first terms, ellipsis, last term with n; don't add*

$2^0 = 1$ and $(-1)^0 = 1$, so the first term is 1. Simplifying $\frac{2}{2}$ to 1 is fine either way.

Extra practice (not scored)

E1. $\left\{(-1)^n \frac{2^n}{2n+1}\right\}$

$(-1)^1\frac{2}{3},\ (-1)^2\frac{4}{5},\ (-1)^3\frac{8}{7},\ (-1)^4\frac{16}{9},\ (-1)^5\frac{32}{11}$ — *plug in $n=1$ to 5*

$\boxed{-\frac{2}{3},\ \frac{4}{5},\ -\frac{8}{7},\ \frac{16}{9},\ -\frac{32}{11}}$ — *odd n negative*

E2. $a_1=1,\ a_2=2,\ a_n=a_{n-1}-a_{n-2}$

$a_3=a_2-a_1=2-1=1$ — *formula starts at $n=3$*

$a_4=a_3-a_2=1-2=-1,\ \ a_5=a_4-a_3=-1-1=-2$

$\boxed{1,\ 2,\ 1,\ -1,\ -2}$

E3. $\frac{3}{4}-\frac{4}{9}+\frac{5}{16}-\cdots+\frac{21}{400}$

top: $k+2$, bottom: $(k+1)^2$ — *tops 3, 4, 5; bottoms $2^2, 3^2, 4^2$*

$+,-,+\ \Rightarrow (-1)^{k+1};\ \ 400=20^2\Rightarrow k=19$ — *last term: $k+1=20$; check top $19+2=21$*

$$\boxed{\sum_{k=1}^{19}(-1)^{k+1}\frac{k+2}{(k+1)^2}}$$